

ENGG 686 – Climate Change Adaptation for Engineers

Exercise 1: Development of a simple climate model and analysis of climate change scenarios

Group Number:

24

Group Members:

Ghazal Sheikhbanouei (UCID : 30269874)

James Inedu (UCID: 30265274)

Name of the TA:

Sharad Aryal

All group members contributed equally to this report and agree with its content.

Date of submission: 2/2/2026

Question 1:

- (a):

This question develops a simple planetary climate model. The model updates the thermal energy stored in the Earth's surface layer from the difference between absorbed solar radiation and outgoing longwave radiation, then converts thermal energy back to an average surface temperature.

The following table shows the parameters given and assumptions:

Table 1 Parameters and Assumptions

Parameter	Value	Units
Solar constant, E_0	1370	$\text{W} \cdot \text{m}^{-2}$
Earth radius, R	6,378,000	m
Earth total surface area, $A = 4\pi R^2$	5.112e+14	m^2
Earth cross-sectional area, $A_x = \pi R^2$	1.278e+14	m^2
Ocean mixed-layer depth, d (base)	100	m
Sea-water density, ρ	1000	$\text{kg} \cdot \text{m}^{-3}$
Specific heat capacity, C_p	4184	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$
Albedo, α (base)	0.30	—
Emissivity, ε (base)	1.0	—
Stefan–Boltzmann constant, σ	5.67×10^{-8}	$\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$
Time step, dt	1	year
Simulation horizon	$t=0 \dots 40$	years
Initial temperature, tempInitial	0	$^{\circ}\text{C}$
Initial temperature, T_0	273	K

Note: Incoming and outgoing energy, E_{in} and E_{out} are computed in Watts ($\text{J} \cdot \text{s}^{-1}$) and converted to annual Joules by multiplying by seconds per year.

Conceptual model

Figure 1 shows the conceptual structure of the simple planetary climate model based on an energy-balance approach. Incoming solar radiation (E_{in}) is partially absorbed by the Earth's surface after accounting for albedo (α). This absorbed shortwave energy increases the thermal energy stored in the ocean mixed layer. At the same time, the Earth loses energy through outgoing longwave radiation (E_{out}), which depends on surface temperature through the Stefan–Boltzmann law.

The net difference between E_{in} and E_{out} determines the change in stored heat energy (E_{Heat}) in the surface layer. This stored heat is related to temperature through the mixed-layer heat capacity term ($A \cdot d \cdot \rho \cdot C_p$), where A is the Earth's surface area, d is the ocean mixed-layer depth, ρ is seawater density, and C_p is the specific heat capacity. As stored heat increases, temperature rises, which in turn increases outgoing radiation, creating a feedback that drives the system toward radiative equilibrium ($E_{in} \approx E_{out}$).

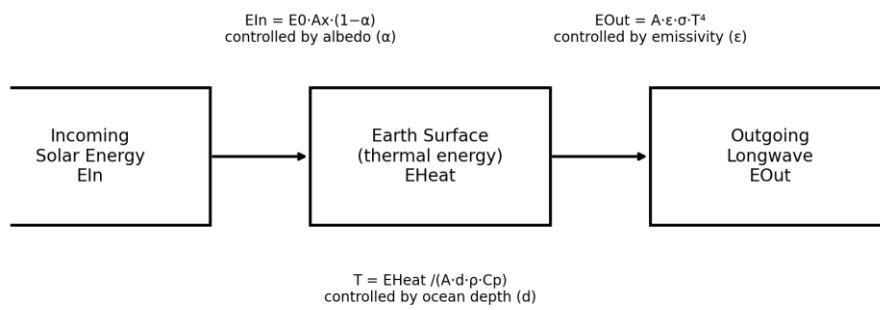


Figure 1: Conceptual structure of the simple climate model

- (b)¹

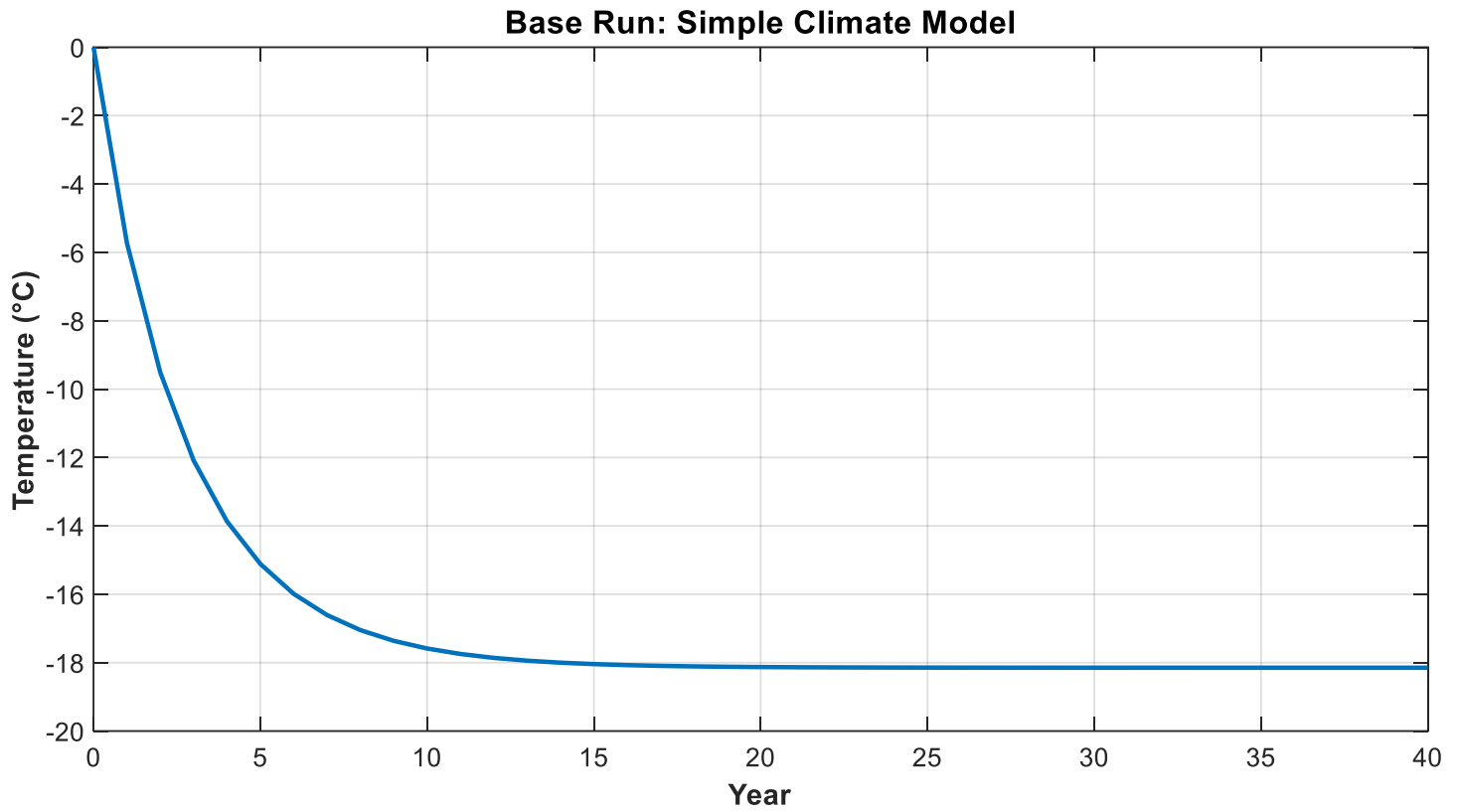


Figure 2: Base-Run Simulation of the Simple Climate Model

Final temperature after 40 years = -18.15 °C

¹ The MATLAB codes used for the simulations are provided in Appendix

- (c)

In the sensitivity experiments, changing each parameter shows how different physical processes influence the climate system. Increasing the ocean mixed-layer depth increases the heat capacity of the system, which slows the rate of temperature change because more energy is required to warm the water. However, the final equilibrium temperature remains the same, showing that depth affects the response time but not the long-term energy balance. In contrast, changing albedo alters how much incoming solar radiation is absorbed. A higher albedo reflects more sunlight and leads to a cooler equilibrium temperature, while a lower albedo results in greater absorption and warming. Similarly, emissivity controls how efficiently the Earth emits outgoing longwave radiation. Higher emissivity increases energy loss and lowers the equilibrium temperature, whereas lower emissivity reduces energy loss and raises the equilibrium temperature. Overall, the results show that ocean depth influences the speed of adjustment, while albedo and emissivity determine the final temperature by modifying the planetary energy balance.

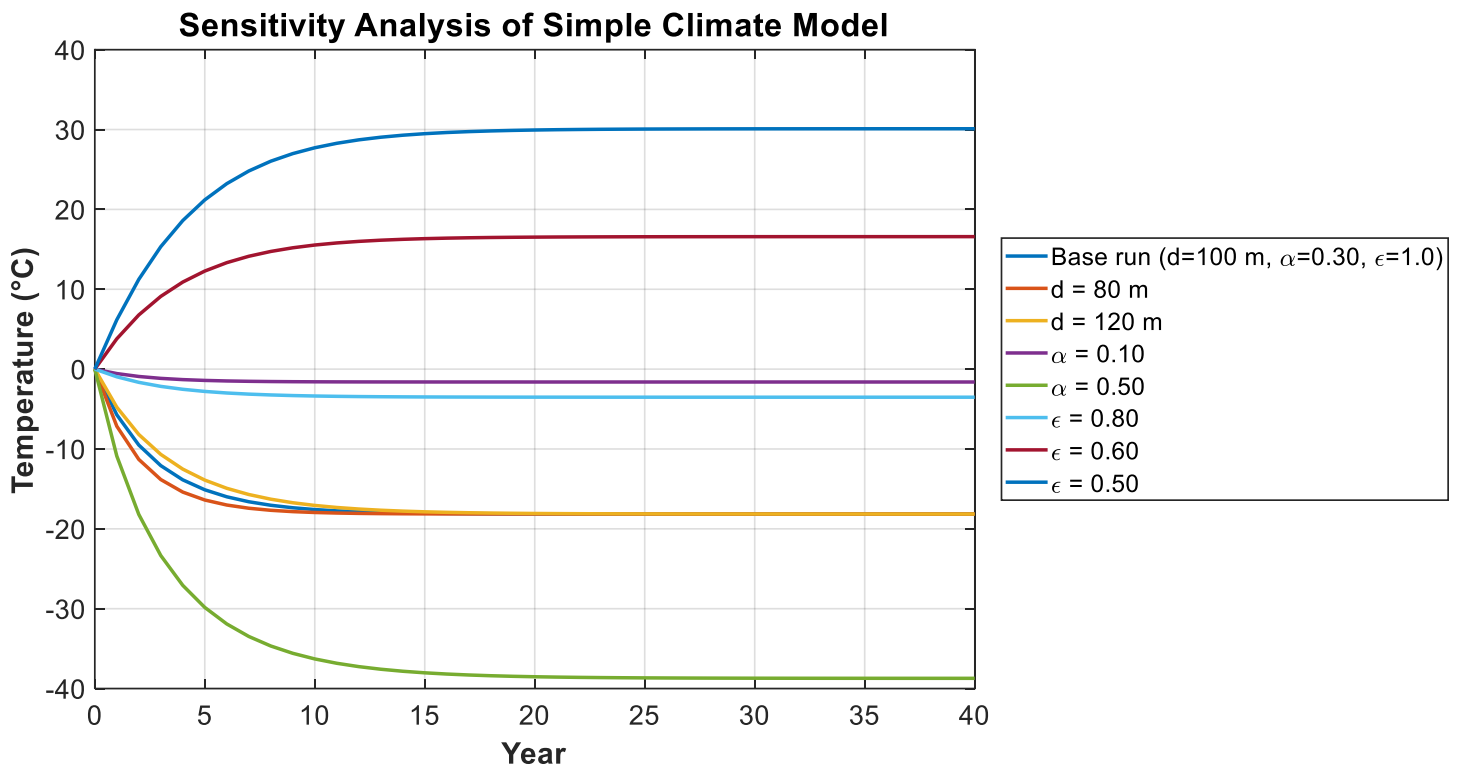


Figure 3: Sensitivity Analysis of the Simple Climate Model

- (d)

Based on the graphical outcomes of the model and the governing physical equations, the variables are categorized as follows regarding their sensitivity:

1. Least Sensitive Variable: Ocean Depth (d)

From a qualitative point of view, Ocean Depth (d) is the least sensitive variable regarding the final state of the climate system.

- **Observation:** In the simulation, varying the ocean depth (e.g., 80m, 100m, 120m) alters the curvature of the temperature plot during the initial years but does not change the final value. All depth scenarios converge to the exact same equilibrium temperature ($\approx -18^{\circ}\text{C}$).
- **Physical Reasoning:** Ocean depth determines the heat capacity of the system. A deeper ocean acts like a larger thermal battery, taking longer to charge or discharge. While this affects the time constant, it has no place in the radiative equilibrium equation. Therefore, it does not dictate the final temperature of the planet.

2. Most Sensitive Variable: Emissivity (ϵ)

Emissivity is found to be the most sensitive variable, particularly concerning the habitability of the planet.

- **Observation:** Changing emissivity produces the most dramatic shift in the results. Reducing ϵ from 1.0 (no atmosphere) to 0.6 (with atmosphere) shifts the temperature from a frozen state ($\approx -18^{\circ}\text{C}$) to a habitable state ($\approx +15^{\circ}\text{C}$). This represents a massive net change of roughly 33°C .
- **Physical Reasoning:** Emissivity represents the efficiency of the Earth in cooling itself. Lower emissivity implies a stronger Greenhouse Effect, where the atmosphere traps outgoing thermal radiation. Because ϵ is a denominator in the energy balance equation, small reductions in it lead to significant increases in surface temperature.

This qualitative assessment is supported by the standard derivation of planetary equilibrium temperature. According to **Hartmann (2016)**, the equilibrium temperature (T_{eq}) is defined by the balance of incoming solar radiation and outgoing terrestrial radiation:

$$T_{eq} = \sqrt[4]{\frac{S_0(1 - \alpha)}{4\sigma\epsilon}}$$

As seen in this governing equation:

1. **ϵ and σ are present:** They directly determine the final temperature (T_{eq})
2. **d (Depth) is absent:** Heat capacity does not appear in the equilibrium formula; it only governs the transient response (the time it takes to reach T_{eq}).

Question 2

- (a)

Part (a) Results Summary (1981-2020)

CO2 Concentration:

RMSE: 3.4036 ppm

Eq: $\text{Obs} = 1.0912 * \text{Model} + -30.9608$

R-squared: 0.9992

Temperature:

RMSE: 0.1892 deg C

Eq: $\text{Obs} = 0.8884 * \text{Model} + 1.4682$

R-squared: 0.8413

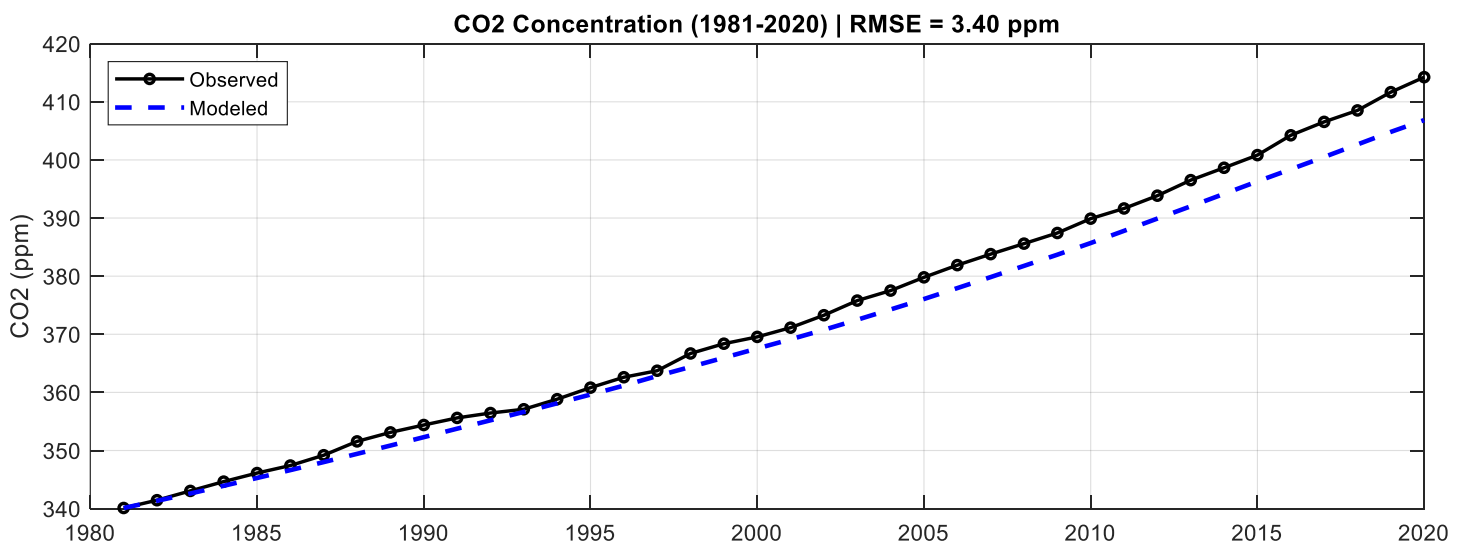


Figure 4: Comparison of observed (black) and modeled (blue) atmospheric CO₂ concentrations from 1981 to 2020.

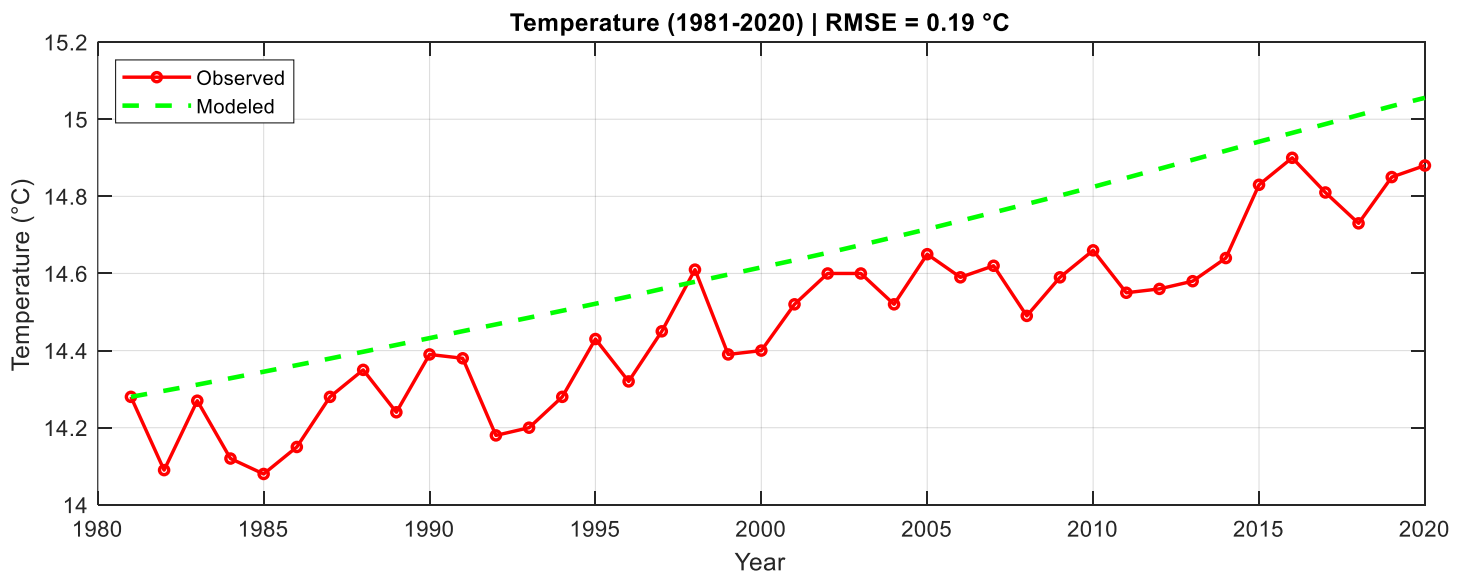


Figure 5: Comparison of observed (red) and modeled (green) global mean temperatures from 1981 to 2020.

- **(b)**

Check emissions values:

2020 (from data): 10.36 GtC/yr

2021-2100 (used): 10.36 GtC/yr

Final modeled values in 2100:

Emissions: 10.36 GtC/yr

CO2 Conc: 569.01 ppm

Temperature: 16.51 °C

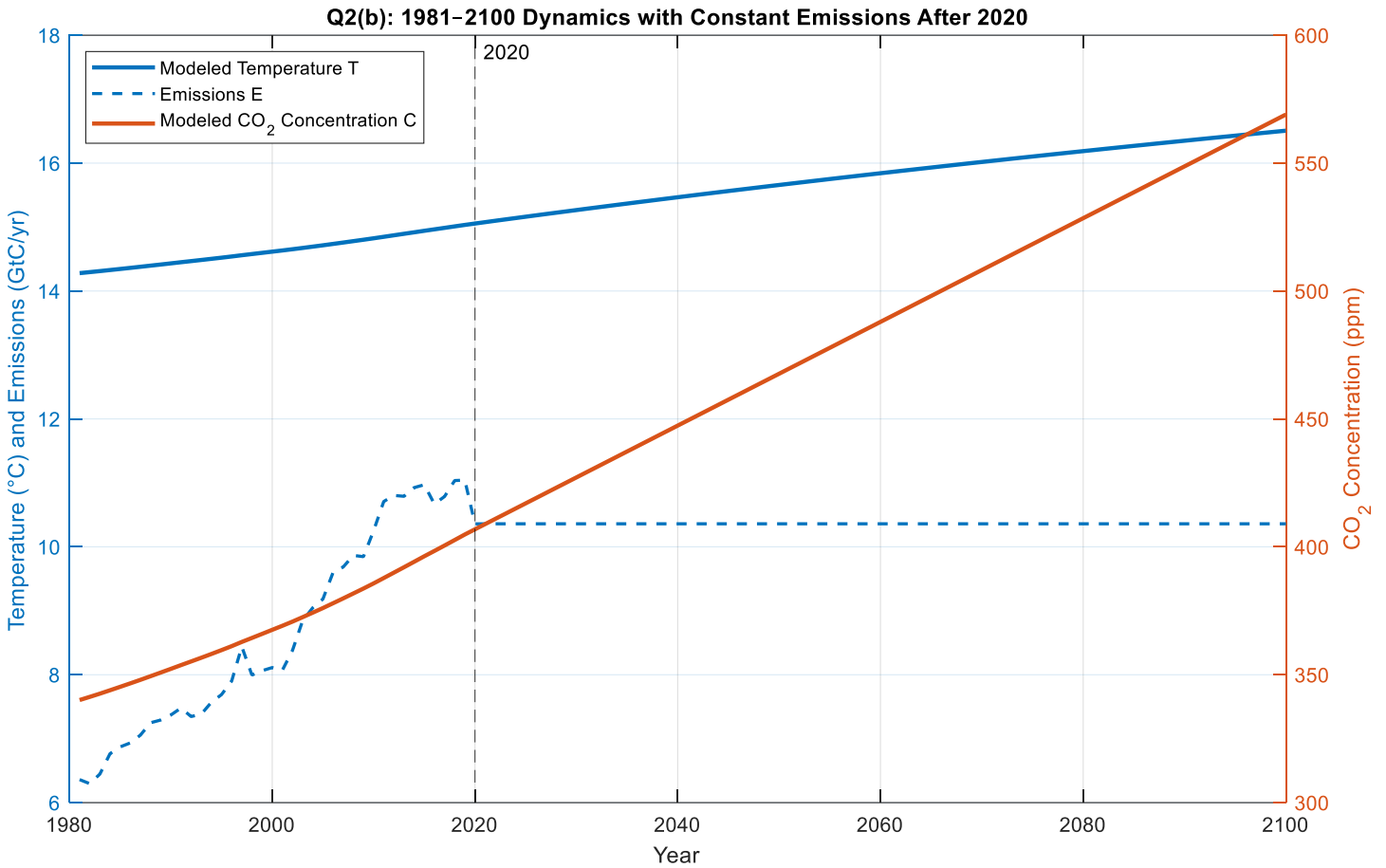


Figure 6: Modeled temperature (T) and atmospheric CO₂ concentration (C) from 1981 to 2100 assuming constant CO₂ emissions at the 2020 level after 2020. Emissions (E) and temperature are shown on the primary axis; CO₂ concentration is shown on the secondary axis.

Even when emissions E are constant from 2021–2100, both CO₂ concentration C and temperature T continue to change. A constant inflow (emissions) still causes the water level (CO₂ concentration) to rise as long as outflow/removal is not equal to inflow. In this simplified model, concentration evolves as

$$C(t) = C(t-1) + E(t) \cdot (0.45/2.3),$$

so if E(t) is a constant positive number, C(t) increases by the same increment every year, producing an approximately linear rise in CO₂ concentration during 2021–2100.

Temperature changes because it depends on the change in radiative forcing driven by CO₂, represented here with a logarithmic relationship:

$$T(t) = T(t-1) + S \cdot \log_2(C(t)/C(t-1)).$$

As CO₂ continues to grow, the ratio C(t)/C(t-1) remains slightly greater than 1 each year, so $\log_2(C(t)/C(t-1))$ is positive, which adds warming annually. However, because the forcing is logarithmic in CO₂, the incremental warming per year tends to be modest and can slowly decrease if the fractional growth of CO₂ becomes smaller over time.

- (c)

Linear regression (1981-2020): $E = 0.131842 \cdot \text{Year} + -255.062829$

Final modeled values in 2100:

Emissions: 21.81 GtC/yr

CO₂ Conc: 666.64 ppm

Temperature: 17.19 °C

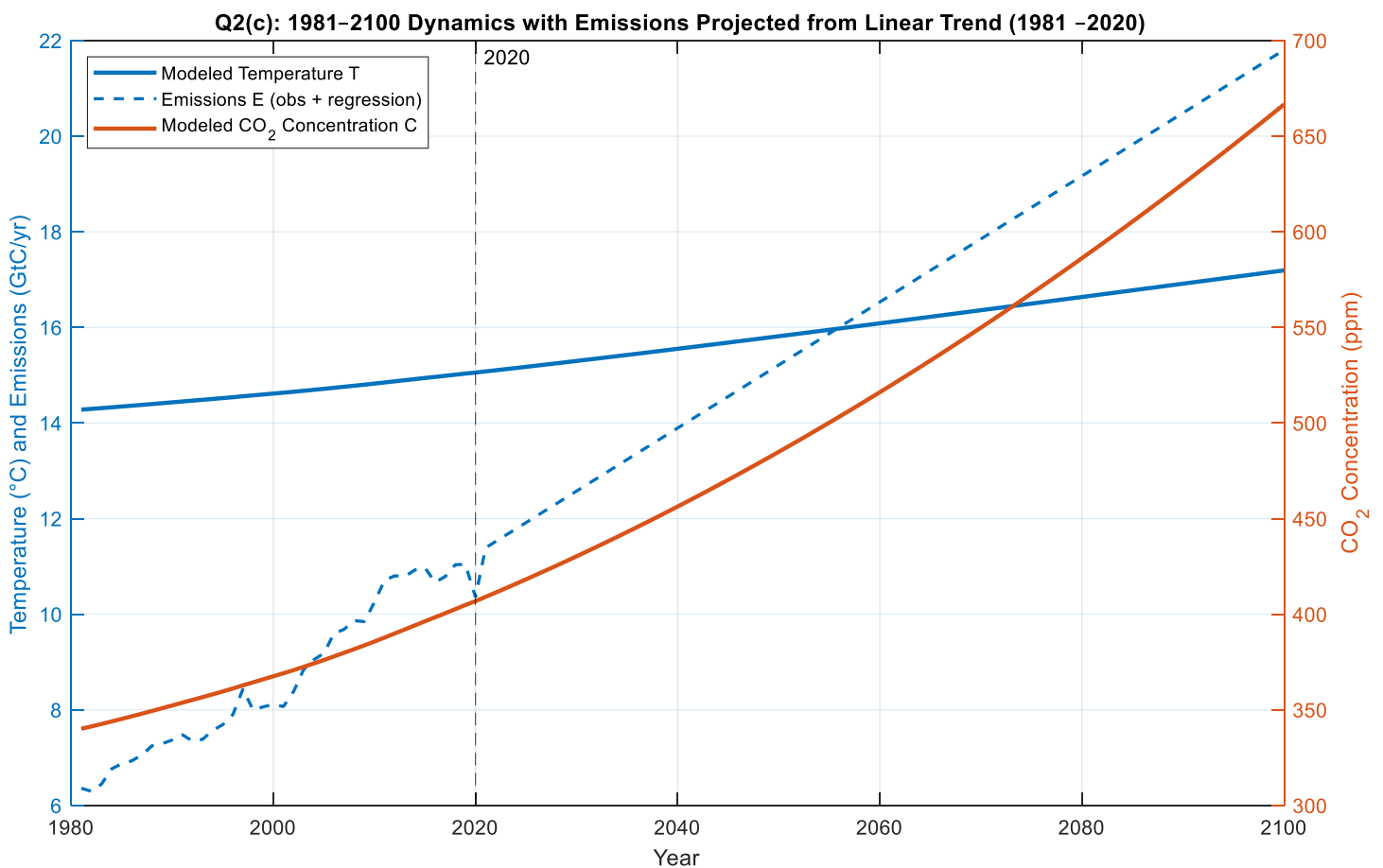


Figure 7: Climate dynamics (1981–2100) with CO₂ emissions projected from a linear regression of 1981–2020 emissions.

In Part (c), future CO₂ emissions are estimated using a linear regression based on historical data from 1981–2020. This means we assume that emissions will continue to change at the same average rate observed in the past. Instead of holding emissions constant, we extend the historical trend forward to predict emissions from 2021 to 2100.

Because emissions continue to increase in this scenario, atmospheric CO₂ concentration also rises more rapidly over time. In the model, concentration evolves according to

$$C(t) = C(t-1) + E(t) \cdot (0.45/2.3).$$

As emissions grow larger each year, more CO₂ is added to the atmosphere annually, causing concentration to increase at an accelerating pace compared to the constant-emissions case.

Temperature rises because it depends on changes in CO₂ concentration through a logarithmic relationship:

$$T(t) = T(t-1) + S \cdot \log_2(C(t)/C(t-1)).$$

Since CO₂ concentration keeps increasing, the ratio $C(t)/C(t-1)$ remains greater than 1, which adds warming each year. Although the warming effect of CO₂ becomes less sensitive at higher concentrations due to the logarithmic form, the continued growth in emissions leads to persistent temperature increases throughout the century.

Overall, assuming a linear trend in emissions results in steadily rising emissions, faster CO₂ accumulation, and greater long-term warming than in the constant-emissions scenario.

- (d)

Regression (1981-2020): $E = 0.131842 \cdot \text{Year} + -255.062829$

Scenario check:

$$E(2050) = 15.21 \text{ GtC/yr}$$

$$E(2051) = 14.91 \text{ GtC/yr (should be } 0.98 * E(2050))$$

Final values in 2100:

Emissions: 5.54 GtC/yr

CO₂ Conc: 577.66 ppm

Temperature: 16.57 °C

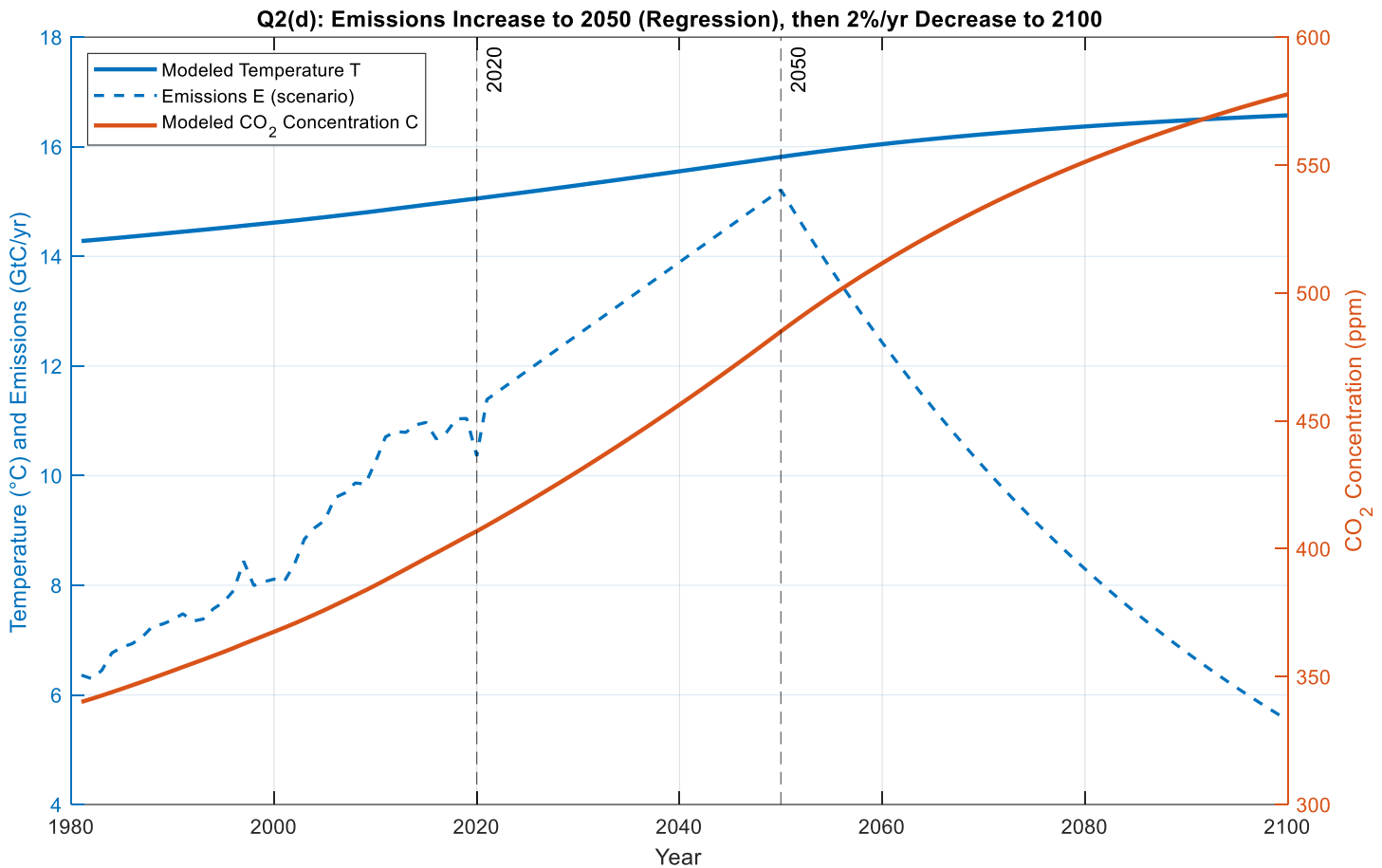


Figure 8: Climate dynamics (1981–2100) with emissions increasing to 2050 (linear-trend projection) and decreasing by 2% per year from 2051–2100.

In Part (d), future CO₂ emissions are assumed to follow the historical linear trend (from Part c) until the year 2050. This means emissions continue increasing at roughly the same average rate observed between 1981 and 2020. After 2050, a mitigation scenario is introduced where emissions decrease by 2% each year until 2100. This creates a pathway where emissions first grow and then gradually decline.

Because emissions are still increasing between 2021 and 2050, atmospheric CO₂ concentration continues to rise more quickly during this period. In the model, concentration depends directly on emissions through the equation $C(t) = C(t-1) + E(t) \cdot (0.45/2.3)$.

Higher emissions add more CO₂ to the atmosphere each year, so concentration keeps building up.

After 2050, even though emissions begin to decrease, CO₂ concentration does not immediately drop. Instead, it continues to rise, but at a slower rate. This is because emissions are still positive, meaning CO₂ is still being added to the atmosphere, just in smaller amounts. Only if emissions were reduced to very low or negative values would concentration start to stabilize or decline.

Temperature keeps increasing throughout most of the century because it depends on changes in CO₂ concentration using a logarithmic relationship. As long as CO₂ levels are rising, the temperature continues to increase. However, after 2050, the rate of warming slows because CO₂ growth slows as well.

Overall, this scenario shows how delayed emission reductions can slow warming but may not be enough to stop continued increases in CO₂ concentration and temperature.

Question 3

Representative Concentration Pathways (RCPs) and implications for adaptation and mitigation

Representative Concentration Pathways (RCPs) are standardized scenarios used in climate modeling that describe possible future trajectories of atmospheric greenhouse-gas concentrations (and other forcing agents). They are not defined by a single socioeconomic factor but by the radiative forcing (the change in Earth's energy balance) which they produce by the year 2100. The RCP family used heavily in the IPCC Fifth Assessment Report comprises four pathways labelled by their approximate forcing in 2100: RCP2.6, RCP4.5, RCP6.0, and RCP8.5, which correspond to ~2.6, 4.5, 6.0 and 8.5 W/m², respectively (Table 3).

Mitigation implication

Because radiative forcing causes long-term warming, moving from a higher to a lower RCP requires stronger mitigation: lower fossil-fuel use, improved efficiency, and/or carbon removal. RCP2.6 represents a strong mitigation (forcing peaks and then declines), whereas RCP8.5 represents continued high emissions with rising forcing. Mitigation planning uses RCPs to test whether proposed policies and technologies can bend forcing toward lower pathways and to estimate the associated reductions in climate risks.

Adaptation implication

Adaptation is about managing impacts under a range of possible futures. RCPs provide a structured way to contain risk: lower pathways generally imply smaller changes in temperature, precipitation extremes, sea level rise and related hazards, while higher pathways imply larger and faster changes. Engineers and planners often use multiple RCPs to stress-test designs (e.g., drainage, culverts, coastal protection, heat-resilient infrastructure) and to evaluate the value of flexibility, safety factors, and staged upgrades. A common practice is to treat a high pathway (historically RCP8.5) as a conservative risk, screen and compare with intermediate pathways (RCP4.5/6.0) to understand the sensitivity of decisions.

In summary, RCPs connect emissions choices to physical climate outcomes through radiative forcing: they are inputs to climate models and a practical framework for evaluating mitigation and designing adaptation.

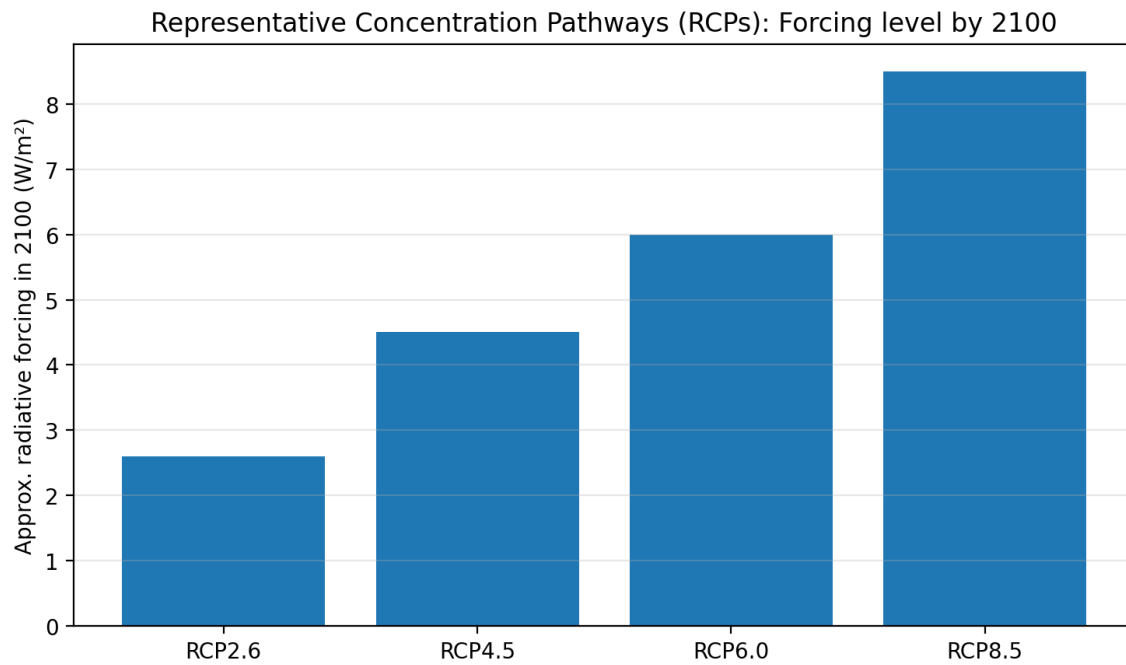


Figure 9: RCP labels correspond to the approximate radiative forcing level in 2100.

Table 2: Summary of the four main RCPs used in AR5

RCP	Forcing in 2100 (W/m ²)	Plain-language interpretation
RCP2.6	2.6	Strong mitigation; forcing peaks ~3 W/m ² mid-century then declines to 2.6 by 2100
RCP4.5	4.5	Stabilization pathway; forcing stabilizes near 4.5 W/m ² after 2100
RCP6.0	6.0	Stabilization pathway; forcing stabilizes near 6.0 W/m ² after 2100
RCP8.5	8.5	High emissions pathway; forcing rises to ~8.5 W/m ² by 2100

Appendix

Question 1/ part b codes:

```
clc; clear; close all;

%% ===== USER-CONTROLLED PLOT SETTINGS =====
FONT_SIZE_AXIS   = 18;    % Axis numbers size
FONT_SIZE_LABEL  = 20;    % X/Y label size
FONT_SIZE_TITLE  = 22;    % Title size
LINE_WIDTH       = 3;     % Plot line thickness
MARKER_SIZE      = 8;     % Marker size (if used)
FIGURE_WIDTH     = 900;   % Figure width (pixels)
FIGURE_HEIGHT    = 600;   % Figure height (pixels)
GRID_LINE_WIDTH  = 1.2;   % Grid line thickness
%% =====

%% Time settings
t = 0:1:40;        % years
dt = 1;            % year
n = length(t);

%% Constants
Eo = 1370;         % Solar constant (W/m^2)
alpha = 0.30;      % Albedo
epsilon = 1.0;     % Emissivity
sigma = 5.67e-8;   % Stefan-Boltzmann (W/m^2/K^4)

d = 100;           % Ocean depth (m)
rho = 1000;        % Density (kg/m^3)
Cp = 4184;         % Heat capacity (J/kg/K)

sec_year = 365*24*60*60; % seconds per year

%% Initial conditions
tempC = zeros(1,n);
tempK = zeros(1,n);
```

```

tempC(1) = 0; % Initial temperature (°C)
tempK(1) = tempC(1) + 273.15;

%% Time loop (energy balance per m^2)
for i = 1:n-1

    % Incoming solar energy (J/m^2/year)
    EIn = (Eo/4) * (1 - alpha) * sec_year;

    % Outgoing longwave radiation (J/m^2/year)
    EOut = epsilon * sigma * tempK(i)^4 * sec_year;

    % Net energy
    dE = EIn - EOut;

    % Temperature change
    dT = dE / (rho * Cp * d);

    % Update temperature
    tempK(i+1) = tempK(i) + dT;
    tempC(i+1) = tempK(i+1) - 273.15;
end

%% ===== PLOT =====
figure('Position',[100 100 FIGURE_WIDTH FIGURE_HEIGHT]);

plot(t, tempC, ...
     'LineWidth', LINE_WIDTH);

grid on;
set(gca, ...
     'FontSize', FONT_SIZE_AXIS, ...
     'LineWidth', GRID_LINE_WIDTH);

xlabel('Year', ...
     'FontSize', FONT_SIZE_LABEL, ...
     'FontWeight', 'bold');

ylabel('Temperature (°C)', ...
     'FontSize', FONT_SIZE_LABEL, ...
     'FontWeight', 'bold');

title('Base Run: Simple Climate Model', ...
     'FontSize', FONT_SIZE_TITLE, ...
     'FontWeight', 'bold');

box on;
%% =====
fprintf('Final temperature after %d years = %.2f °C\n', t(end), tempC(end));

```

Question 1/ part c codes:

```

%% ===== USER-CONTROLLED PLOT SETTINGS =====
FONT_SIZE_AXIS = 20;
FONT_SIZE_LABEL = 22;
FONT_SIZE_TITLE = 24;
LINE_WIDTH = 2.5;
FIG_W = 1000;
FIG_H = 650;
%% =====

%% Time settings
t = 0:1:40; % years
n = length(t);
sec_year = 365*24*60*60;

%% Constants
Eo = 1370; % Solar constant (W/m^2)
sigma = 5.67e-8; % Stefan-Boltzmann constant
rho = 1000; % Density (kg/m^3)
Cp = 4184; % Heat capacity (J/kg/K)

```

```

%% Base parameters
d_base      = 100;
alpha_base  = 0.30;
epsilon_base = 1.0;

%% Sensitivity values
d_values    = [80 120];
alpha_values = [0.10 0.50];
epsilon_values = [0.80 0.60 0.50];

%% Storage for plotting
results = {};
labels  = {};

%% ===== FUNCTION =====
climate_model = @(d,alpha,epsilon) ...
    ( ...
        function_handle(d,alpha,epsilon) ...
    );

%% ===== BASE RUN =====
tempC = zeros(1,n);
tempK = zeros(1,n);
tempK(1) = 273.15;

for i = 1:n-1
    EIn = (Eo/4) * (1 - alpha_base) * sec_year;
    EOut = epsilon_base * sigma * tempK(i)^4 * sec_year;
    dT = (EIn - EOut) / (rho * Cp * d_base);
    tempK(i+1) = tempK(i) + dT;
    tempC(i+1) = tempK(i+1) - 273.15;
end

results{end+1} = tempC;
labels{end+1} = 'Base run (d=100 m, \alpha=0.30, \epsilon=1.0)';

%% ===== DEPTH VARIATION =====
for d = d_values
    tempK = 273.15 * ones(1,n);
    tempC = zeros(1,n);

    for i = 1:n-1
        EIn = (Eo/4) * (1 - alpha_base) * sec_year;
        EOut = epsilon_base * sigma * tempK(i)^4 * sec_year;
        dT = (EIn - EOut) / (rho * Cp * d);
        tempK(i+1) = tempK(i) + dT;
        tempC(i+1) = tempK(i+1) - 273.15;
    end

    results{end+1} = tempC;
    labels{end+1} = sprintf('d = %d m', d);
end

%% ===== ALBEDO VARIATION =====
for alpha = alpha_values
    tempK = 273.15 * ones(1,n);
    tempC = zeros(1,n);

    for i = 1:n-1
        EIn = (Eo/4) * (1 - alpha) * sec_year;
        EOut = epsilon_base * sigma * tempK(i)^4 * sec_year;
        dT = (EIn - EOut) / (rho * Cp * d_base);
        tempK(i+1) = tempK(i) + dT;
        tempC(i+1) = tempK(i+1) - 273.15;
    end

    results{end+1} = tempC;
    labels{end+1} = sprintf('\alpha = %.2f', alpha);
end

%% ===== EMISSIVITY VARIATION =====
for epsilon = epsilon_values
    tempK = 273.15 * ones(1,n);

```

```

tempC = zeros(1,n);

for i = 1:n-1
    EIn = (Eo/4) * (1 - alpha_base) * sec_year;
    EOut = epsilon * sigma * tempK(i)^4 * sec_year;
    dT = (EIn - EOut) / (rho * Cp * d_base);
    tempK(i+1) = tempK(i) + dT;
    tempC(i+1) = tempK(i+1) - 273.15;
end

results{end+1} = tempC;
labels{end+1} = sprintf('\epsilon = %.2f', epsilon);
end

%% ===== PLOT =====
figure('Position',[100 100 FIG_W FIG_H]); hold on;

for i = 1:length(results)
    plot(t, results{i}, 'LineWidth', LINE_WIDTH);
end

grid on;
set(gca, 'FontSize', FONT_SIZE_AXIS, 'LineWidth', 1.2);

xlabel('Year', 'FontSize', FONT_SIZE_LABEL, 'FontWeight', 'bold');
ylabel('Temperature (°C)', 'FontSize', FONT_SIZE_LABEL, 'FontWeight', 'bold');
title('Sensitivity Analysis of Simple Climate Model', 'FontSize', FONT_SIZE_TITLE, 'FontWeight', 'bold');

legend(labels, 'Location', 'eastoutside');
box on;

```

Question 2/ part a codes:

```

clc;
clear;
close all;

%% 1. Load Data
% Make sure the CSV file is in your Current Folder
filename = 'Exercice 01_data for Question 2.xlsx';

% Check if file exists
if ~isfile(filename)
    error('File not found! Please ensure "%s" is in your current MATLAB folder.', filename);
end

data = readtable(filename);

% Extract columns (Using column indices is safer if headers have spaces)
% Column 1: Year, Col 3: E (GtC), Col 4: C_obs, Col 5: T_obs
Year = data(:, 1);
E_GtC = data(:, 3);
C_obs = data(:, 4);
T_obs = data(:, 5);

%% 2. Initialize Model
n = length(Year);
C_model = zeros(n, 1);
T_model = zeros(n, 1);

% Initial Conditions (1981 is index 1)
C_model(1) = C_obs(1);
T_model(1) = T_obs(1);

% Parameters
S = 3; % Climate Sensitivity
factor = 0.45 / 2.3; % Conversion factor for Emissions

%% 3. Simulation Loop (1982 to 2020)
for t = 2:n
    % Calculate CO2 Concentration: C(t) = C(t-1) + E(t)*0.45/2.3
    C_model(t) = C_model(t-1) + (E_GtC(t) * factor);

```

```

    % Calculate Temperature:  $T(t) = T(t-1) + S * \log_2(C(t)/C(t-1))$ 
    ratio = C_model(t) / C_model(t-1);
    T_model(t) = T_model(t-1) + S * log2(ratio);
end

%% 4. Statistical Analysis

% Root Mean Square Error (RMSE)
rmse_C = sqrt(mean((C_obs - C_model).^2));
rmse_T = sqrt(mean((T_obs - T_model).^2));

% Linear Regression: Observed vs Modeled
% fitlm is part of the Statistics Toolbox. If not available, use polyfit.

% For CO2
p_C = polyfit(C_model, C_obs, 1); % p(1) is slope, p(2) is intercept
slope_C = p_C(1);
int_C = p_C(2);
yfit_C = polyval(p_C, C_model);
% R-squared Calculation
SStot_C = sum((C_obs - mean(C_obs)).^2);
SSres_C = sum((C_obs - yfit_C).^2);
r2_C = 1 - (SSres_C / SStot_C);

% For Temperature
p_T = polyfit(T_model, T_obs, 1);
slope_T = p_T(1);
int_T = p_T(2);
yfit_T = polyval(p_T, T_model);
% R-squared Calculation
SStot_T = sum((T_obs - mean(T_obs)).^2);
SSres_T = sum((T_obs - yfit_T).^2);
r2_T = 1 - (SSres_T / SStot_T);

% Display Results
fprintf('-----\n');
fprintf('Part (a) Results Summary (1981-2020)\n');
fprintf('-----\n');
fprintf('CO2 Concentration:\n');
fprintf(' RMSE:      %.4f ppm\n', rmse_C);
fprintf(' Eq:        Obs = %.4f * Model + %.4f\n', slope_C, int_C);
fprintf(' R-squared: %.4f\n', r2_C);
fprintf('\n');
fprintf('Temperature:\n');
fprintf(' RMSE:      %.4f deg C\n', rmse_T);
fprintf(' Eq:        Obs = %.4f * Model + %.4f\n', slope_T, int_T);
fprintf(' R-squared: %.4f\n', r2_T);
fprintf('-----\n');

%% 5. Plotting
figure('Position', [100, 100, 1000, 800]);

% Subplot 1: CO2
subplot(2, 1, 1);
plot(Year, C_obs, 'ko-', 'LineWidth', 1.5, 'MarkerSize', 4); hold on;
plot(Year, C_model, 'b--', 'LineWidth', 2);
title(['CO2 Concentration (1981-2020) | RMSE = ' num2str(rmse_C, '%.2f') ' ppm']);
ylabel('CO2 (ppm)');
legend('Observed', 'Modeled', 'Location', 'NorthWest');
grid on;

% Subplot 2: Temperature
subplot(2, 1, 2);
plot(Year, T_obs, 'ro-', 'LineWidth', 1.5, 'MarkerSize', 4); hold on;
plot(Year, T_model, 'g--', 'LineWidth', 2);
title(['Temperature (1981-2020) | RMSE = ' num2str(rmse_T, '%.2f') ' °C']);
ylabel('Temperature (°C)');
xlabel('Year');
legend('Observed', 'Modeled', 'Location', 'NorthWest');
grid on;

% Optional: Save the plot
saveas(gcf, 'Q2a_Comparison_MATLAB.png');

```

Question 2/ part b codes:

```
%% Q2(b) - Very Simple Climate Model
% Constant CO2 emissions at 2020 level (10.36 GtC/yr) for 2021-2100
% Plot 1981-2100 with:
%   left y-axis: T and E
%   right y-axis: C

clear; clc; close all;

%% 1) Load Historical Data (1981-2020)
filename = 'Exercice 01_data for Question 2.xlsx';

% If your data is on a specific sheet, uncomment and adjust:
% data = readtable(filename, 'Sheet', 'Sheet2');
data = readtable(filename);

% Extract historical arrays (assuming columns match your earlier code)
hist_year = data(:, 1); % Year (1981-2020)
hist_E = data(:, 3); % Emissions (GtC/yr)
hist_C = data(:, 4); % CO2 concentration (ppm)
hist_T = data(:, 5); % Temperature (°C)

%% 2) Constants
S = 3; % climate sensitivity parameter
conversion_factor = 0.45 / 2.3; % ppm increase per GtC emitted

%% 3) Setup Simulation (1981-2100)
years_all = (1981:2100)';
n_total = length(years_all);
n_hist = length(hist_year);

E_sim = nan(n_total, 1);
C_sim = nan(n_total, 1);
T_sim = nan(n_total, 1);

% Initial conditions from 1981 observed values
C_sim(1) = hist_C(1);
T_sim(1) = hist_T(1);

% Fill historical emissions 1981-2020
E_sim(1:n_hist) = hist_E;

% Emissions constant after 2020 (use 2020 value from data OR force 10.36)
E2020_data = hist_E(end); % emission in 2020 from your dataset
E2020 = 10.36; % given by question (GtC/yr)

% Use the value specified by the question:
E_sim(n_hist+1:end) = E2020;

%% 4) Run Simulation Loop (1982-2100)
for i = 2:n_total
    % CO2 concentration
    C_sim(i) = C_sim(i-1) + (E_sim(i) * conversion_factor);

    % Temperature
    ratio = C_sim(i) / C_sim(i-1);
    T_sim(i) = T_sim(i-1) + S * log2(ratio);
end

%% 5) Plot (Dual Y-Axis)
figure('Position', [100, 100, 1000, 600]);

yyaxis left
plot(years_all, T_sim, 'LineWidth', 2); hold on;
plot(years_all, E_sim, '--', 'LineWidth', 1.5);
ylabel('Temperature (°C) and Emissions (GtC/yr)');

yyaxis right
plot(years_all, C_sim, 'LineWidth', 2);
ylabel('CO2 Concentration (ppm)');
```

```

xlabel('Year');
title('Q2(b): 1981-2100 Dynamics with Constant Emissions After 2020');
grid on;

xline(2020, '--k', '2020', 'LabelOrientation', 'horizontal');

legend('Modeled Temperature T', 'Emissions E', 'Modeled CO_2 Concentration C', ...
'Location', 'northwest');

%% 6) Print key values
fprintf('Check emissions values:\n');
fprintf(' 2020 (from data): %.2f GtC/yr\n', E2020_data);
fprintf(' 2021-2100 (used): %.2f GtC/yr\n\n', E2020);

fprintf('Final modeled values in 2100:\n');
fprintf(' Emissions:   %.2f GtC/yr\n', E_sim(end));
fprintf(' CO2 Conc:    %.2f ppm\n', C_sim(end));
fprintf(' Temperature: %.2f °C\n', T_sim(end));

```

Question 2 / part c codes:

```

%% Q2(c) - Linear regression of E vs Year (1981-2020), then forecast E (2021-2100)
% Then model C and T using:
%   C(t) = C(t-1) + E(t)*0.45/2.3
%   T(t) = T(t-1) + S*log2( C(t)/C(t-1) )

clear; clc; close all;

%% 1) Load Historical Data (1981-2020)
filename = 'Excercise 01_data for Question 2.xlsx';

% If needed (recommended if multiple sheets):
% data = readtable(filename, 'Sheet', 'Sheet2');
data = readtable(filename);

hist_year = data(:, 1); % Year
hist_E    = data(:, 3); % Emissions (GtC/yr)
hist_C    = data(:, 4); % CO2 concentration (ppm)
hist_T    = data(:, 5); % Temperature (°C)

%% 2) Constants
S = 3;
conversion_factor = 0.45 / 2.3;

%% 3) Linear regression: E vs year for 1981-2020
% E = a*year + b
p = polyfit(hist_year, hist_E, 1);
a = p(1);
b = p(2);

fprintf('Linear regression (1981-2020): E = %.6f*Year + %.6f\n', a, b);

%% 4) Setup Simulation (1981-2100)
years_all = (1981:2100)';
n_total   = length(years_all);
n_hist    = length(hist_year);

E_sim = nan(n_total, 1);
C_sim = nan(n_total, 1);
T_sim = nan(n_total, 1);

% Initial conditions (1981 observed)
C_sim(1) = hist_C(1);
T_sim(1) = hist_T(1);

% Fill historical E
E_sim(1:n_hist) = hist_E;

% Forecast E for 2021-2100 using regression equation
for i = (n_hist+1):n_total
    E_sim(i) = polyval(p, years_all(i));
end

```

```

% Optional safety: prevent negative emissions if regression extrapolates badly
E_sim(E_sim < 0) = 0;

%% 5) Run model forward (1982-2100)
for i = 2:n_total
    C_sim(i) = C_sim(i-1) + (E_sim(i) * conversion_factor);

    ratio = C_sim(i) / C_sim(i-1);
    T_sim(i) = T_sim(i-1) + S * log2(ratio);
end

%% 6) Plot dynamics (1981-2100): T & E left, C right
figure('Position', [100, 100, 1000, 600]);

yyaxis left
plot(years_all, T_sim, 'LineWidth', 2); hold on;
plot(years_all, E_sim, '--', 'LineWidth', 1.5);
ylabel('Temperature (°C) and Emissions (GtC/yr)');

yyaxis right
plot(years_all, C_sim, 'LineWidth', 2);
ylabel('CO2 Concentration (ppm)');

xlabel('Year');
title('Q2(c): 1981-2100 Dynamics with Emissions Projected from Linear Trend (1981-2020)');
grid on;

xline(2020, '--k', '2020', 'LabelOrientation', 'horizontal');

legend('Modeled Temperature T', 'Emissions E (obs + regression)', 'Modeled CO2 Concentration C', ...
    'Location', 'northwest');

%% 7) Print end values
fprintf('\nFinal modeled values in 2100:\n');
fprintf(' Emissions:   %.2f GtC/yr\n', E_sim(end));
fprintf(' CO2 Conc:     %.2f ppm\n', C_sim(end));
fprintf(' Temperature: %.2f °C\n', T_sim(end));

```

Question 2/ part d codes:

```

%% Q2(d) - Use regression from (c) to 2050, then reduce E by 2%/yr to 2100
% Equations:
%   C(t) = C(t-1) + E(t)*0.45/2.3
%   T(t) = T(t-1) + S*log2( C(t)/C(t-1) )

clear; clc; close all;

%% 1) Load Historical Data (1981-2020)
filename = 'Exercice 01_data for Question 2.xlsx';

% If needed (recommended if multiple sheets):
% data = readtable(filename, 'Sheet', 'Sheet2');
data = readtable(filename);

hist_year = data(:, 1); % Year
hist_E    = data(:, 3); % Emissions (GtC/yr)
hist_C    = data(:, 4); % CO2 concentration (ppm)
hist_T    = data(:, 5); % Temperature (°C)

%% 2) Constants
S = 3;
conversion_factor = 0.45 / 2.3;

%% 3) Regression of E vs Year (1981-2020): E = a*Year + b
p = polyfit(hist_year, hist_E, 1); % p(1)=a, p(2)=b
a = p(1); b = p(2);
fprintf('Regression (1981-2020): E = %.6f*Year + %.6f\n', a, b);

%% 4) Setup Simulation (1981-2100)
years_all = (1981:2100)';
n_total   = length(years_all);
n_hist     = length(hist_year);

```

```

E_sim = nan(n_total, 1);
C_sim = nan(n_total, 1);
T_sim = nan(n_total, 1);

% Initial conditions from 1981 observed
C_sim(1) = hist_C(1);
T_sim(1) = hist_T(1);

% Fill observed emissions (1981-2020)
E_sim(1:n_hist) = hist_E;

%% 5) Build emissions scenario for 2021-2100 (Part d)
% 2021-2050: use regression equation (continue increasing)
% 2051-2100: reduce E by 2% every year (E = previous * 0.98)

for i = (n_hist+1):n_total
    y = years_all(i);

    if y <= 2050
        E_sim(i) = polyval(p, y); % regression projection
    else
        E_sim(i) = E_sim(i-1) * 0.98; % 2% annual reduction
    end
end

% Optional safety: emissions cannot be negative
E_sim(E_sim < 0) = 0;

%% 6) Run climate model forward (1982-2100)
for i = 2:n_total
    C_sim(i) = C_sim(i-1) + (E_sim(i) * conversion_factor);

    ratio = C_sim(i) / C_sim(i-1);
    T_sim(i) = T_sim(i-1) + S * log2(ratio);
end

%% 7) Plot dynamics (1981-2100): T & E left axis, C right axis
figure('Position', [100, 100, 1000, 600]);

yyaxis left
plot(years_all, T_sim, 'LineWidth', 2); hold on;
plot(years_all, E_sim, '--', 'LineWidth', 1.5);
ylabel('Temperature (°C) and Emissions (GtC/yr)');

yyaxis right
plot(years_all, C_sim, 'LineWidth', 2);
ylabel('CO_2 Concentration (ppm)');

xlabel('Year');
title('Q2(d): Emissions Increase to 2050 (Regression), then 2%/yr Decrease to 2100');
grid on;

xline(2020, '--k', '2020');
xline(2050, '--k', '2050');

legend('Modeled Temperature T', 'Emissions E (scenario)', 'Modeled CO_2 Concentration C', ...
    'Location', 'northwest');

%% 8) Print key values
idx2050 = find(years_all == 2050, 1);
fprintf('\nScenario check:\n');
fprintf(' E(2050) = %.2f GtC/yr\n', E_sim(idx2050));
fprintf(' E(2051) = %.2f GtC/yr (should be 0.98 * E2050)\n', E_sim(idx2050+1));
fprintf('\nFinal values in 2100:\n');
fprintf(' Emissions: %.2f GtC/yr\n', E_sim(end));
fprintf(' CO2 Conc: %.2f ppm\n', C_sim(end));
fprintf(' Temperature: %.2f °C\n', T_sim(end));

```

References

- Bralower, T., & Bice, D. (n.d.). *Introduction to a simple planetary climate model (EARTH 103)*. Penn State University.
- Center for Science Education, UCAR. (n.d.). *The very simple climate model*. University Corporation for Atmospheric Research.
- Government of Canada. (n.d.). *Representative Concentration Pathways (RCPs)*. Canadian Centre for Climate Services.
- Hartmann, D. L. (2016). *Global physical climatology* (2nd ed., pp. 25–27). Elsevier.

- IPCC. (2013). *Climate change 2013: The physical science basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- van Vuuren, D. P., Edmonds, J., Kainuma, M., Riahi, K., Thomson, A., Hibbard, K., Hurtt, G. C., Kram, T., Krey, V., Lamarque, J.-F., Masui, T., Meinshausen, M., Nakicenovic, N., Smith, S. J., & Rose, S. K. (2011). The representative concentration pathways: An overview. *Climatic Change*, 109(1–2), 5–31.
<https://doi.org/10.1007/s10584-011-0148-z>